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MIL-F-47076A (MI)
24 September 1990
SUPERSEDING
MIL-F-47076 (MI)
10 May 1974

MILITARY SPECIFICATION

FILM, POLYETHYLENE TEREPHTHALATE, METALLIZED

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for aluminum metallized polyethylene terephthalate film.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8135

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STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-171	-	Finishing of Metal and Wood Surfaces

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material. The material shall consist of a polyethylene terephthalate film having one side coated with 99.5 percent pure aluminum deposited by the vacuum deposition technique.

3.2.1 Base film material. The base film material, including thickness, shall be specified by the procuring activity (see 6.2) and shall conform to the requirements of table I.

TABLE I. Base film material.

Base film	Specification	Thickness, Inches (Nominal)
Polyethylene Terephthalate	MIL-I-631, Type G, Form F, Class I	0.001 to .0075

3.2.2 Protective coating. When specified by the procuring activity (see 6.2), the metallized side of film shall have a conversion coating applied in conformance with MIL-STD-171, Finish Number 7.3.

3.3 Physical property.

3.3.1 Aluminum thickness. The thickness of aluminum shall be 1 to 2 micro inches.

3.4 Workmanship. The metallized film shall show no blisters, bloom or tear holes. The metallic coating shall be bright, uniform, clean and free from bare spots, scratches, holes and longitudinal striations (see 4.7).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.

4.4 First article inspection. When specified (see 6.2) a sample shall be subjected to first article inspection in accordance with all examinations and tests specified herein.

4.5 Quality conformance inspection. Quality conformance inspection for acceptance of material shall consist of all examinations, tests and inspection requirements of this specification and shall be performed by the supplier on a sample of metallized material representative of the shipment. A certified test report shall be furnished by the supplier and shall include the results of all examinations, tests and inspections. The Government reserves the right to conduct any or all of the examinations and inspections prescribed herein on each lot of metallized material submitted by the supplier.

4.5.1 Lot formulation. Lot size shall consist of all the material submitted for acceptance at the same time, which has been prepared by the same manufacturer with no change in materials or processes and in one continuous period of operation.

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4.5.1.1 Sampling. Unless specified in the contract or order, sampling for inspection or conformance shall be as specified in table II.

TABLE II. Sampling plan.

Lot size	Sample size
2 to 3	All
4 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 and over	29
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.6 Test methods.

4.6.1 Physical property test.

4.6.1.1 Aluminum thickness. The aluminum thickness shall be determined by inspecting the optical density of the metallized film.

4.6.1.1.1 Test equipment. A Welsh Densichron instrument, or equivalent, with a blue sensitive probe and a fluorescent light source are required to check optical density.

4.6.1.1.2 Test procedure. The Densichron amplifier shall be preset to read zero through unmetallized film. The metallized film shall then be tested at intervals across the width and length as specified in table III. The metallized film thickness shall be considered within the limits specified in 3.3.1 providing the optical density reads 2.5 $\pm$ 0.5.

TABLE III. Optical density inspection areas.

Width of film	Inspection area, distance from edge of film	Inspection intervals, maximum
1-12 inches	0.50 of width	100 feet
12-24 inches	0.33 and 0.67 of width	100 feet
24 inches and over	0.25, 0.50 and 0.75 of width	100 feet

4.6.2 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-STD-129.

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4.7 Examinations. The film shall be visually examined to assure conformance to the workmanship requirements specified in 3.4.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified, packaging and packing shall be Level C.

5.1.1 Level C packaging. Material shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity.

5.1.2 Level C packing. Packages shall be secured and packed in shipping containers in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material covered by this specification may be used as a vapor transmission barrier. Other applications should be substantiated by testing for the properties in question.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Whether a first article sample is required (see 3.1)

d. Base film material requirement (see 3.2.1)

e. Base film material thickness requirement (see 3.2.1)

f. Whether protective coating is required (see 3.2.2)

g. Where first article sample test shall be performed (see 4.4)

h. Request for supplier certification (see 4.5)

i. Sampling plan, if other than specified (see 4.5.1.1)

j. Any special marking requirements (see 5.1).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Aluminum shield
Protective coating
Vapor barrier

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Project No. 8135-A626

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-47076A(MI)	

3. DOCUMENT TITLE
FILM, POLYETHYLENE TEREPHTHALATE, METALIZED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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